

实验报告

作业1

Version 1

补充代码内容、将损失函数修改为交叉熵损失函数，得到单个 hidden layer 效果：

```
Epoch 1 Loss: 0.9162, Acc: 0.7756: 100%|██████████| 782/782 [00:03<00:00, 223.79it/s]
Epoch 1 Val Acc: 0.8864
Epoch 2 Loss: 0.4451, Acc: 0.8815: 100%|██████████| 782/782 [00:03<00:00, 223.43it/s]
Epoch 2 Val Acc: 0.9038
Epoch 3 Loss: 0.3734, Acc: 0.8974: 100%|██████████| 782/782 [00:03<00:00, 239.96it/s]
Epoch 3 Val Acc: 0.9114
Epoch 4 Loss: 0.3383, Acc: 0.9055: 100%|██████████| 782/782 [00:03<00:00, 232.99it/s]
Epoch 4 Val Acc: 0.9172
Epoch 5 Loss: 0.3149, Acc: 0.9111: 100%|██████████| 782/782 [00:03<00:00, 213.41it/s]
Epoch 5 Val Acc: 0.9217
Epoch 6 Loss: 0.2966, Acc: 0.9171: 100%|██████████| 782/782 [00:03<00:00, 228.46it/s]
Epoch 6 Val Acc: 0.9241
Epoch 7 Loss: 0.2821, Acc: 0.9209: 100%|██████████| 782/782 [00:03<00:00, 246.77it/s]
Epoch 7 Val Acc: 0.9287
Epoch 8 Loss: 0.2686, Acc: 0.9249: 100%|██████████| 782/782 [00:03<00:00, 232.77it/s]
Epoch 8 Val Acc: 0.9315
Epoch 9 Loss: 0.2573, Acc: 0.9287: 100%|██████████| 782/782 [00:03<00:00, 219.22it/s]
Epoch 9 Val Acc: 0.9339
Epoch 10 Loss: 0.2467, Acc: 0.9311: 100%|██████████| 782/782 [00:03<00:00, 239.77it/s]
Epoch 10 Val Acc: 0.9361
Final Test Accuracy: 0.9335
```

分析：交叉熵损失函数更适合多分类任务，相比于原始的均方误差，交叉熵加快了收敛速度。

Version 2

调整网络结构，使用两个 hidden layer，分别 256 个神经元：

```
Epoch 1 Loss: 0.7913, Acc: 0.8017: 100%|██████████| 782/782 [00:05<00:00, 141.29it/s]
Epoch 1 Val Acc: 0.9006
Epoch 2 Loss: 0.3508, Acc: 0.9019: 100%|██████████| 782/782 [00:05<00:00, 147.66it/s]
Epoch 2 Val Acc: 0.9200
Epoch 3 Loss: 0.2917, Acc: 0.9178: 100%|██████████| 782/782 [00:04<00:00, 158.39it/s]
Epoch 3 Val Acc: 0.9261
Epoch 4 Loss: 0.2583, Acc: 0.9269: 100%|██████████| 782/782 [00:04<00:00, 160.32it/s]
Epoch 4 Val Acc: 0.9364
Epoch 5 Loss: 0.2349, Acc: 0.9332: 100%|██████████| 782/782 [00:04<00:00, 177.55it/s]
Epoch 5 Val Acc: 0.9403
Epoch 6 Loss: 0.2161, Acc: 0.9385: 100%|██████████| 782/782 [00:04<00:00, 165.23it/s]
Epoch 6 Val Acc: 0.9456
Epoch 7 Loss: 0.2004, Acc: 0.9434: 100%|██████████| 782/782 [00:04<00:00, 172.45it/s]
Epoch 7 Val Acc: 0.9494
Epoch 8 Loss: 0.1870, Acc: 0.9466: 100%|██████████| 782/782 [00:04<00:00, 170.79it/s]
Epoch 8 Val Acc: 0.9507
Epoch 9 Loss: 0.1754, Acc: 0.9496: 100%|██████████| 782/782 [00:04<00:00, 168.04it/s]
Epoch 9 Val Acc: 0.9543
Epoch 10 Loss: 0.1645, Acc: 0.9528: 100%|██████████| 782/782 [00:04<00:00, 171.82it/s]
Epoch 10 Val Acc: 0.9569
Final Test Accuracy: 0.9528
```

分析：将网络结构加深为两层隐藏层，提高了模型对特征的非线性表达能力和参数容量，最终测试集准确率稳定在95%以上。

Version 3

修改激活函数，分别使用 ReLu 和 tanh（第一层 ReLU，第二层 Tanh）：

```
Epoch 1 Loss: 0.7333, Acc: 0.8143: 100%|██████████| 782/782 [00:06<00:00, 127.15it/s]
Epoch 1 Val Acc: 0.9043
Epoch 2 Loss: 0.3515, Acc: 0.9026: 100%|██████████| 782/782 [00:05<00:00, 137.21it/s]
Epoch 2 Val Acc: 0.9215
Epoch 3 Loss: 0.2892, Acc: 0.9182: 100%|██████████| 782/782 [00:05<00:00, 140.28it/s]
Epoch 3 Val Acc: 0.9323
Epoch 4 Loss: 0.2537, Acc: 0.9277: 100%|██████████| 782/782 [00:05<00:00, 148.00it/s]
Epoch 4 Val Acc: 0.9385
Epoch 5 Loss: 0.2285, Acc: 0.9345: 100%|██████████| 782/782 [00:05<00:00, 143.80it/s]
Epoch 5 Val Acc: 0.9435
Epoch 6 Loss: 0.2083, Acc: 0.9400: 100%|██████████| 782/782 [00:05<00:00, 150.82it/s]
Epoch 6 Val Acc: 0.9482
Epoch 7 Loss: 0.1919, Acc: 0.9450: 100%|██████████| 782/782 [00:05<00:00, 146.97it/s]
Epoch 7 Val Acc: 0.9517
Epoch 8 Loss: 0.1782, Acc: 0.9491: 100%|██████████| 782/782 [00:05<00:00, 147.86it/s]
Epoch 8 Val Acc: 0.9551
Epoch 9 Loss: 0.1666, Acc: 0.9524: 100%|██████████| 782/782 [00:05<00:00, 149.77it/s]
Epoch 9 Val Acc: 0.9571
Epoch 10 Loss: 0.1563, Acc: 0.9558: 100%|██████████| 782/782 [00:05<00:00, 148.90it/s]
Epoch 10 Val Acc: 0.9578
Final Test Accuracy: 0.9548
```

分析：意外发现混合激活函数的效果极佳。第一层采用 ReLU 提取稀疏化特征且保证梯度顺利下传，第二层采用 tanh 将特征数据零中心化并平滑映射到 (-1,1) 区间，这种特性的互补使得最后一层分类边界更加稳定。

Version 4

都使用 tanh：

```

Epoch 1 Loss: 0.6606, Acc: 0.8260: 100%|██████████| 782/782 [00:05<00:00, 134.93it/s]
Epoch 1 Val Acc: 0.9008
Epoch 2 Loss: 0.3674, Acc: 0.8973: 100%|██████████| 782/782 [00:05<00:00, 140.79it/s]
Epoch 2 Val Acc: 0.9139
Epoch 3 Loss: 0.3191, Acc: 0.9097: 100%|██████████| 782/782 [00:05<00:00, 133.68it/s]
Epoch 3 Val Acc: 0.9188
Epoch 4 Loss: 0.2924, Acc: 0.9174: 100%|██████████| 782/782 [00:05<00:00, 141.21it/s]
Epoch 4 Val Acc: 0.9247
Epoch 5 Loss: 0.2720, Acc: 0.9226: 100%|██████████| 782/782 [00:05<00:00, 142.23it/s]
Epoch 5 Val Acc: 0.9284
Epoch 6 Loss: 0.2561, Acc: 0.9271: 100%|██████████| 782/782 [00:05<00:00, 141.61it/s]
Epoch 6 Val Acc: 0.9313
Epoch 7 Loss: 0.2421, Acc: 0.9309: 100%|██████████| 782/782 [00:05<00:00, 140.90it/s]
Epoch 7 Val Acc: 0.9349
Epoch 8 Loss: 0.2296, Acc: 0.9349: 100%|██████████| 782/782 [00:05<00:00, 142.19it/s]
Epoch 8 Val Acc: 0.9386
Epoch 9 Loss: 0.2180, Acc: 0.9376: 100%|██████████| 782/782 [00:06<00:00, 125.13it/s]
Epoch 9 Val Acc: 0.9417
Epoch 10 Loss: 0.2078, Acc: 0.9408: 100%|██████████| 782/782 [00:05<00:00, 137.21it/s]
Epoch 10 Val Acc: 0.9452
Final Test Accuracy: 0.9420

```

分析：效果不如纯 ReLU。因为 tanh 在输入偏大时会进入两侧饱和区，导数趋于 0。连续多层使用会导致反向传播时的梯度消失，初期收敛非常缓慢。

Version 5

都使用 Sigmoid：

```

Epoch 1 Loss: 2.2310, Acc: 0.2823: 100%|██████████| 782/782 [00:06<00:00, 127.79it/s]
Epoch 1 Val Acc: 0.4715
Epoch 2 Loss: 2.0148, Acc: 0.5578: 100%|██████████| 782/782 [00:05<00:00, 137.59it/s]
Epoch 2 Val Acc: 0.6655
Epoch 3 Loss: 1.6640, Acc: 0.6740: 100%|██████████| 782/782 [00:05<00:00, 150.11it/s]
Epoch 3 Val Acc: 0.7253
Epoch 4 Loss: 1.2697, Acc: 0.7414: 100%|██████████| 782/782 [00:05<00:00, 150.88it/s]
Epoch 4 Val Acc: 0.7611
Epoch 5 Loss: 0.9828, Acc: 0.7895: 100%|██████████| 782/782 [00:05<00:00, 153.78it/s]
Epoch 5 Val Acc: 0.8264
Epoch 6 Loss: 0.8073, Acc: 0.8172: 100%|██████████| 782/782 [00:05<00:00, 142.22it/s]
Epoch 6 Val Acc: 0.8534
Epoch 7 Loss: 0.6960, Acc: 0.8356: 100%|██████████| 782/782 [00:05<00:00, 147.98it/s]
Epoch 7 Val Acc: 0.8608
Epoch 8 Loss: 0.6204, Acc: 0.8466: 100%|██████████| 782/782 [00:05<00:00, 151.65it/s]
Epoch 8 Val Acc: 0.8703
Epoch 9 Loss: 0.5662, Acc: 0.8571: 100%|██████████| 782/782 [00:05<00:00, 147.61it/s]
Epoch 9 Val Acc: 0.8785
Epoch 10 Loss: 0.5258, Acc: 0.8650: 100%|██████████| 782/782 [00:05<00:00, 144.09it/s]
Epoch 10 Val Acc: 0.8819
Final Test Accuracy: 0.8763

```

分析：表现最差且收敛极慢。Sigmoid 的导数最大仅为 0.25，连乘会引发极其严重的梯度消失；且其输出恒为正数（非零中心化），导致权重更新方向受限，网络前半部分几乎无法有效学习。

最终采用 Version 2

作业2

Version 0

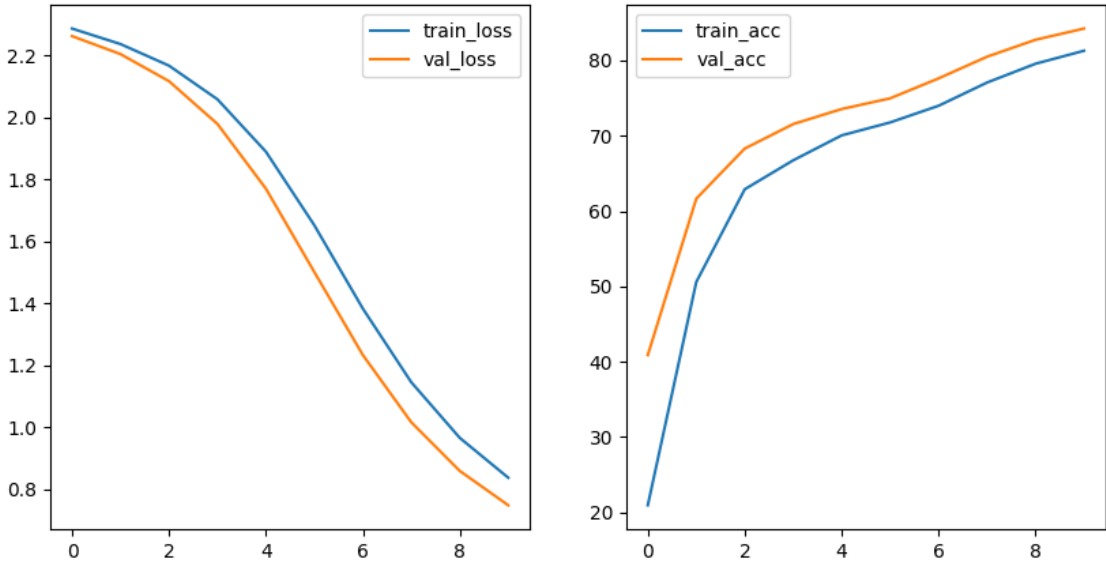
运行 template 代码 · 初始状态结果：

```
Train Epoch: 0 [0/50000 (0%)]    Loss: 2.301843
Train Epoch: 0 [6400/50000 (13%)]    Loss: 2.306332
Train Epoch: 0 [12800/50000 (26%)]    Loss: 2.291170
Train Epoch: 0 [19200/50000 (38%)]    Loss: 2.290346
Train Epoch: 0 [25600/50000 (51%)]    Loss: 2.286119
Train Epoch: 0 [32000/50000 (64%)]    Loss: 2.280914
Train Epoch: 0 [38400/50000 (77%)]    Loss: 2.269774
Train Epoch: 0 [44800/50000 (90%)]    Loss: 2.273836
Train set: Average loss: 2.2861, Accuracy: 10460/50000 (20.92%)
Validation set: Average loss: 2.2616, Accuracy: 4088/10000 (40.88%)
Train Epoch: 1 [0/50000 (0%)]    Loss: 2.262681
Train Epoch: 1 [6400/50000 (13%)]    Loss: 2.254794
Train Epoch: 1 [12800/50000 (26%)]    Loss: 2.247004
Train Epoch: 1 [19200/50000 (38%)]    Loss: 2.228260
Train Epoch: 1 [25600/50000 (51%)]    Loss: 2.232014
Train Epoch: 1 [32000/50000 (64%)]    Loss: 2.227493
Train Epoch: 1 [38400/50000 (77%)]    Loss: 2.217177
Train Epoch: 1 [44800/50000 (90%)]    Loss: 2.227712
Train set: Average loss: 2.2360, Accuracy: 25289/50000 (50.58%)
Validation set: Average loss: 2.2037, Accuracy: 6165/10000 (61.65%)
Train Epoch: 2 [0/50000 (0%)]    Loss: 2.207645
Train Epoch: 2 [6400/50000 (13%)]    Loss: 2.202989
Train Epoch: 2 [12800/50000 (26%)]    Loss: 2.162946
Train Epoch: 2 [19200/50000 (38%)]    Loss: 2.195642
Train Epoch: 2 [25600/50000 (51%)]    Loss: 2.128262
Train Epoch: 2 [32000/50000 (64%)]    Loss: 2.129099
Train Epoch: 2 [38400/50000 (77%)]    Loss: 2.132071
Train Epoch: 2 [44800/50000 (90%)]    Loss: 2.126675
Train set: Average loss: 2.1665, Accuracy: 31436/50000 (62.87%)
Validation set: Average loss: 2.1165, Accuracy: 6831/10000 (68.31%)
Train Epoch: 3 [0/50000 (0%)]    Loss: 2.138174
Train Epoch: 3 [6400/50000 (13%)]    Loss: 2.100654
Train Epoch: 3 [12800/50000 (26%)]    Loss: 2.106171
Train Epoch: 3 [19200/50000 (38%)]    Loss: 2.061829
Train Epoch: 3 [25600/50000 (51%)]    Loss: 2.060261
Train Epoch: 3 [32000/50000 (64%)]    Loss: 2.031607
Train Epoch: 3 [38400/50000 (77%)]    Loss: 2.047221
Train Epoch: 3 [44800/50000 (90%)]    Loss: 2.008823
Train set: Average loss: 2.0580, Accuracy: 33373/50000 (66.75%)
Validation set: Average loss: 1.9786, Accuracy: 7156/10000 (71.56%)
Train Epoch: 4 [0/50000 (0%)]    Loss: 1.984720
Train Epoch: 4 [6400/50000 (13%)]    Loss: 1.996149
```



```
Train Epoch: 4 [12800/50000 (26%)]      Loss: 1.904217
Train Epoch: 4 [19200/50000 (38%)]      Loss: 1.895297
Train Epoch: 4 [25600/50000 (51%)]      Loss: 1.850187
Train Epoch: 4 [32000/50000 (64%)]      Loss: 1.844752
Train Epoch: 4 [38400/50000 (77%)]      Loss: 1.900312
Train Epoch: 4 [44800/50000 (90%)]      Loss: 1.802253
Train set: Average loss: 1.8891, Accuracy: 35037/50000 (70.07%)
Validation set: Average loss: 1.7698, Accuracy: 7356/10000 (73.56%)
Train Epoch: 5 [0/50000 (0%)]      Loss: 1.774061
Train Epoch: 5 [6400/50000 (13%)]      Loss: 1.784344
Train Epoch: 5 [12800/50000 (26%)]      Loss: 1.700992
Train Epoch: 5 [19200/50000 (38%)]      Loss: 1.736155
Train Epoch: 5 [25600/50000 (51%)]      Loss: 1.644886
Train Epoch: 5 [32000/50000 (64%)]      Loss: 1.555481
Train Epoch: 5 [38400/50000 (77%)]      Loss: 1.551002
Train Epoch: 5 [44800/50000 (90%)]      Loss: 1.589711
Train set: Average loss: 1.6516, Accuracy: 35882/50000 (71.76%)
Validation set: Average loss: 1.5009, Accuracy: 7498/10000 (74.98%)
Train Epoch: 6 [0/50000 (0%)]      Loss: 1.434148
Train Epoch: 6 [6400/50000 (13%)]      Loss: 1.592053
Train Epoch: 6 [12800/50000 (26%)]      Loss: 1.468890
Train Epoch: 6 [19200/50000 (38%)]      Loss: 1.316622
Train Epoch: 6 [25600/50000 (51%)]      Loss: 1.315486
Train Epoch: 6 [32000/50000 (64%)]      Loss: 1.347599
Train Epoch: 6 [38400/50000 (77%)]      Loss: 1.264090
Train Epoch: 6 [44800/50000 (90%)]      Loss: 1.358835
Train set: Average loss: 1.3832, Accuracy: 37002/50000 (74.00%)
Validation set: Average loss: 1.2336, Accuracy: 7763/10000 (77.63%)
Train Epoch: 7 [0/50000 (0%)]      Loss: 1.267892
Train Epoch: 7 [6400/50000 (13%)]      Loss: 1.313682
Train Epoch: 7 [12800/50000 (26%)]      Loss: 1.161118
Train Epoch: 7 [19200/50000 (38%)]      Loss: 1.153583
Train Epoch: 7 [25600/50000 (51%)]      Loss: 1.208150
Train Epoch: 7 [32000/50000 (64%)]      Loss: 1.140679
Train Epoch: 7 [38400/50000 (77%)]      Loss: 1.133171
Train Epoch: 7 [44800/50000 (90%)]      Loss: 1.043400
Train set: Average loss: 1.1457, Accuracy: 38541/50000 (77.08%)
Validation set: Average loss: 1.0166, Accuracy: 8051/10000 (80.51%)
Train Epoch: 8 [0/50000 (0%)]      Loss: 1.040685
Train Epoch: 8 [6400/50000 (13%)]      Loss: 0.947586
Train Epoch: 8 [12800/50000 (26%)]      Loss: 1.100791
Train Epoch: 8 [19200/50000 (38%)]      Loss: 0.897734
Train Epoch: 8 [25600/50000 (51%)]      Loss: 1.105176
Train Epoch: 8 [32000/50000 (64%)]      Loss: 0.896132
Train Epoch: 8 [38400/50000 (77%)]      Loss: 0.913312
Train Epoch: 8 [44800/50000 (90%)]      Loss: 0.802273
```

```
Train Epoch: 8 [44800/50000 (90%)]    Loss: 0.892272
Train set: Average loss: 0.9664, Accuracy: 39787/50000 (79.57%)
Validation set: Average loss: 0.8599, Accuracy: 8276/10000 (82.76%)
Train Epoch: 9 [0/50000 (0%)]    Loss: 0.906881
Train Epoch: 9 [6400/50000 (13%)]    Loss: 0.763005
Train Epoch: 9 [12800/50000 (26%)]    Loss: 0.929164
Train Epoch: 9 [19200/50000 (38%)]    Loss: 0.893307
Train Epoch: 9 [25600/50000 (51%)]    Loss: 0.848675
Train Epoch: 9 [32000/50000 (64%)]    Loss: 0.898491
Train Epoch: 9 [38400/50000 (77%)]    Loss: 0.860972
Train Epoch: 9 [44800/50000 (90%)]    Loss: 0.621474
Train set: Average loss: 0.8379, Accuracy: 40657/50000 (81.31%)
Validation set: Average loss: 0.7491, Accuracy: 8425/10000 (84.25%)
Test set: Accuracy: 8309/10000 (83.09%)
```



分析：采用 800 -> 800 -> 10 的两层隐藏层结构，即便随机初始化没有额外技巧，因为隐藏单元数量多，模型也能较快收敛。

Version 1

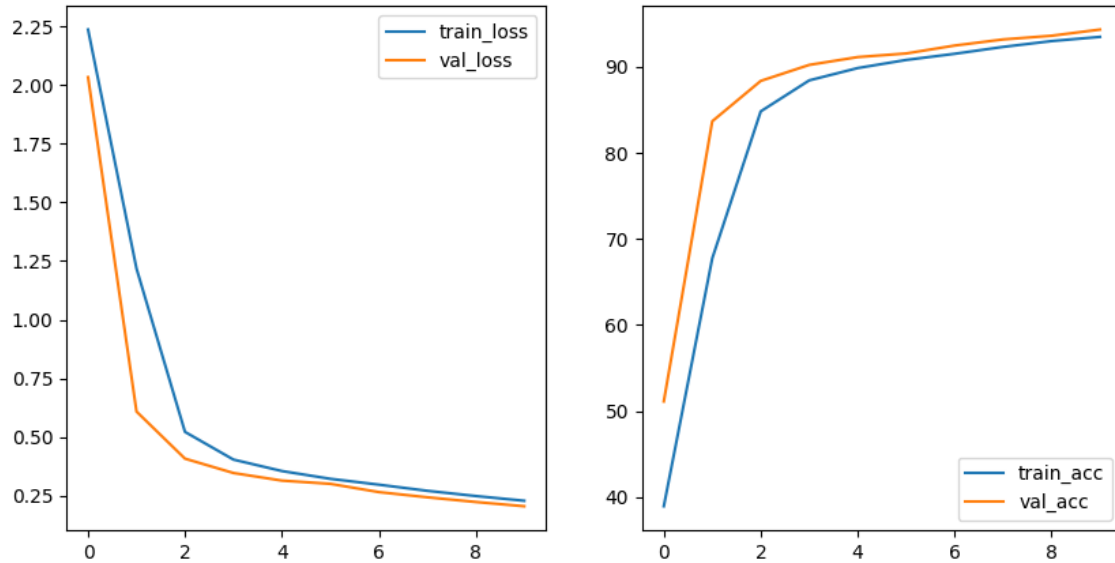
增加隐藏层，调整神经元数量：

```
Train Epoch: 0 [0/50000 (0%)]    Loss: 2.314700
Train Epoch: 0 [6400/50000 (13%)]    Loss: 2.302047
Train Epoch: 0 [12800/50000 (26%)]    Loss: 2.282936
Train Epoch: 0 [19200/50000 (38%)]    Loss: 2.275383
Train Epoch: 0 [25600/50000 (51%)]    Loss: 2.264126
Train Epoch: 0 [32000/50000 (64%)]    Loss: 2.247886
Train Epoch: 0 [38400/50000 (77%)]    Loss: 2.167603
Train Epoch: 0 [44800/50000 (90%)]    Loss: 2.176425
Train set: Average loss: 2.2362, Accuracy: 19454/50000 (38.91%)
Validation set: Average loss: 2.0231, Accuracy: 5111/10000 (51.11%)
```

```
Validation set: Average loss: 2.0331, Accuracy: 5111/10000 (51.11%)
Train Epoch: 1 [0/50000 (0%)] Loss: 2.009359
Train Epoch: 1 [6400/50000 (13%)] Loss: 1.952518
Train Epoch: 1 [12800/50000 (26%)] Loss: 1.629619
Train Epoch: 1 [19200/50000 (38%)] Loss: 1.409466
Train Epoch: 1 [25600/50000 (51%)] Loss: 1.134242
Train Epoch: 1 [32000/50000 (64%)] Loss: 1.048672
Train Epoch: 1 [38400/50000 (77%)] Loss: 0.599829
Train Epoch: 1 [44800/50000 (90%)] Loss: 0.826613
Train set: Average loss: 1.2181, Accuracy: 33860/50000 (67.72%)
Validation set: Average loss: 0.6090, Accuracy: 8367/10000 (83.67%)
Train Epoch: 2 [0/50000 (0%)] Loss: 0.670694
Train Epoch: 2 [6400/50000 (13%)] Loss: 0.563120
Train Epoch: 2 [12800/50000 (26%)] Loss: 0.642045
Train Epoch: 2 [19200/50000 (38%)] Loss: 0.518791
Train Epoch: 2 [25600/50000 (51%)] Loss: 0.775217
Train Epoch: 2 [32000/50000 (64%)] Loss: 0.470131
Train Epoch: 2 [38400/50000 (77%)] Loss: 0.453805
Train Epoch: 2 [44800/50000 (90%)] Loss: 0.429060
Train set: Average loss: 0.5224, Accuracy: 42409/50000 (84.82%)
Validation set: Average loss: 0.4084, Accuracy: 8836/10000 (88.36%)
Train Epoch: 3 [0/50000 (0%)] Loss: 0.356375
Train Epoch: 3 [6400/50000 (13%)] Loss: 0.637775
Train Epoch: 3 [12800/50000 (26%)] Loss: 0.582279
Train Epoch: 3 [19200/50000 (38%)] Loss: 0.513757
Train Epoch: 3 [25600/50000 (51%)] Loss: 0.429409
Train Epoch: 3 [32000/50000 (64%)] Loss: 0.361248
Train Epoch: 3 [38400/50000 (77%)] Loss: 0.347694
Train Epoch: 3 [44800/50000 (90%)] Loss: 0.253403
Train set: Average loss: 0.4043, Accuracy: 44208/50000 (88.42%)
Validation set: Average loss: 0.3472, Accuracy: 9022/10000 (90.22%)
Train Epoch: 4 [0/50000 (0%)] Loss: 0.486445
Train Epoch: 4 [6400/50000 (13%)] Loss: 0.376107
Train Epoch: 4 [12800/50000 (26%)] Loss: 0.185306
Train Epoch: 4 [19200/50000 (38%)] Loss: 0.412222
Train Epoch: 4 [25600/50000 (51%)] Loss: 0.335005
Train Epoch: 4 [32000/50000 (64%)] Loss: 0.328931
Train Epoch: 4 [38400/50000 (77%)] Loss: 0.293475
Train Epoch: 4 [44800/50000 (90%)] Loss: 0.315962
Train set: Average loss: 0.3555, Accuracy: 44927/50000 (89.85%)
Validation set: Average loss: 0.3148, Accuracy: 9113/10000 (91.13%)
Train Epoch: 5 [0/50000 (0%)] Loss: 0.293547
Train Epoch: 5 [6400/50000 (13%)] Loss: 0.401479
Train Epoch: 5 [12800/50000 (26%)] Loss: 0.377090
Train Epoch: 5 [19200/50000 (38%)] Loss: 0.136154
Train Epoch: 5 [25600/50000 (51%)] Loss: 0.508272
```



```
Train Epoch: 5 [32000/50000 (64%)]      Loss: 0.338974
Train Epoch: 5 [38400/50000 (77%)]      Loss: 0.436747
Train Epoch: 5 [44800/50000 (90%)]      Loss: 0.202563
Train set: Average loss: 0.3224, Accuracy: 45400/50000 (90.80%)
Validation set: Average loss: 0.3010, Accuracy: 9155/10000 (91.55%)
Train Epoch: 6 [0/50000 (0%)]      Loss: 0.180267
Train Epoch: 6 [6400/50000 (13%)]      Loss: 0.279093
Train Epoch: 6 [12800/50000 (26%)]     Loss: 0.212186
Train Epoch: 6 [19200/50000 (38%)]     Loss: 0.151106
Train Epoch: 6 [25600/50000 (51%)]     Loss: 0.434351
Train Epoch: 6 [32000/50000 (64%)]     Loss: 0.289709
Train Epoch: 6 [38400/50000 (77%)]     Loss: 0.469952
Train Epoch: 6 [44800/50000 (90%)]     Loss: 0.189334
Train set: Average loss: 0.2975, Accuracy: 45773/50000 (91.55%)
Validation set: Average loss: 0.2659, Accuracy: 9247/10000 (92.47%)
Train Epoch: 7 [0/50000 (0%)]      Loss: 0.249548
Train Epoch: 7 [6400/50000 (13%)]      Loss: 0.227514
Train Epoch: 7 [12800/50000 (26%)]     Loss: 0.189763
Train Epoch: 7 [19200/50000 (38%)]     Loss: 0.182476
Train Epoch: 7 [25600/50000 (51%)]     Loss: 0.198150
Train Epoch: 7 [32000/50000 (64%)]     Loss: 0.243403
Train Epoch: 7 [38400/50000 (77%)]     Loss: 0.130635
Train Epoch: 7 [44800/50000 (90%)]     Loss: 0.197248
Train set: Average loss: 0.2713, Accuracy: 46153/50000 (92.31%)
Validation set: Average loss: 0.2438, Accuracy: 9318/10000 (93.18%)
Train Epoch: 8 [0/50000 (0%)]      Loss: 0.290659
Train Epoch: 8 [6400/50000 (13%)]      Loss: 0.165886
Train Epoch: 8 [12800/50000 (26%)]     Loss: 0.391490
Train Epoch: 8 [19200/50000 (38%)]     Loss: 0.328203
Train Epoch: 8 [25600/50000 (51%)]     Loss: 0.174479
Train Epoch: 8 [32000/50000 (64%)]     Loss: 0.342842
Train Epoch: 8 [38400/50000 (77%)]     Loss: 0.208550
Train Epoch: 8 [44800/50000 (90%)]     Loss: 0.482821
Train set: Average loss: 0.2491, Accuracy: 46491/50000 (92.98%)
Validation set: Average loss: 0.2237, Accuracy: 9361/10000 (93.61%)
Train Epoch: 9 [0/50000 (0%)]      Loss: 0.262088
Train Epoch: 9 [6400/50000 (13%)]      Loss: 0.178264
Train Epoch: 9 [12800/50000 (26%)]     Loss: 0.083921
Train Epoch: 9 [19200/50000 (38%)]     Loss: 0.198857
Train Epoch: 9 [25600/50000 (51%)]     Loss: 0.165265
Train Epoch: 9 [32000/50000 (64%)]     Loss: 0.330503
Train Epoch: 9 [38400/50000 (77%)]     Loss: 0.257901
Train Epoch: 9 [44800/50000 (90%)]     Loss: 0.150351
Train set: Average loss: 0.2290, Accuracy: 46736/50000 (93.47%)
Validation set: Average loss: 0.2056, Accuracy: 9433/10000 (94.33%)
Test set: Accuracy: 9385/10000 (93.85%)
```

分析：将模型调整为更深的 512 -> 256 -> 128 -> 10 结构后，整体效果有了进一步的提升。对比原本极为宽浅的结构，逐层降维的深层网络迫使模型在每层向前传递时不断浓缩信息，提取出更加高级、抽象的非线性特征。这种多层级的特征提取机制极大地增强了模型的表达能力和特征整合能力，使其在验证集上的表现变得更为优秀。

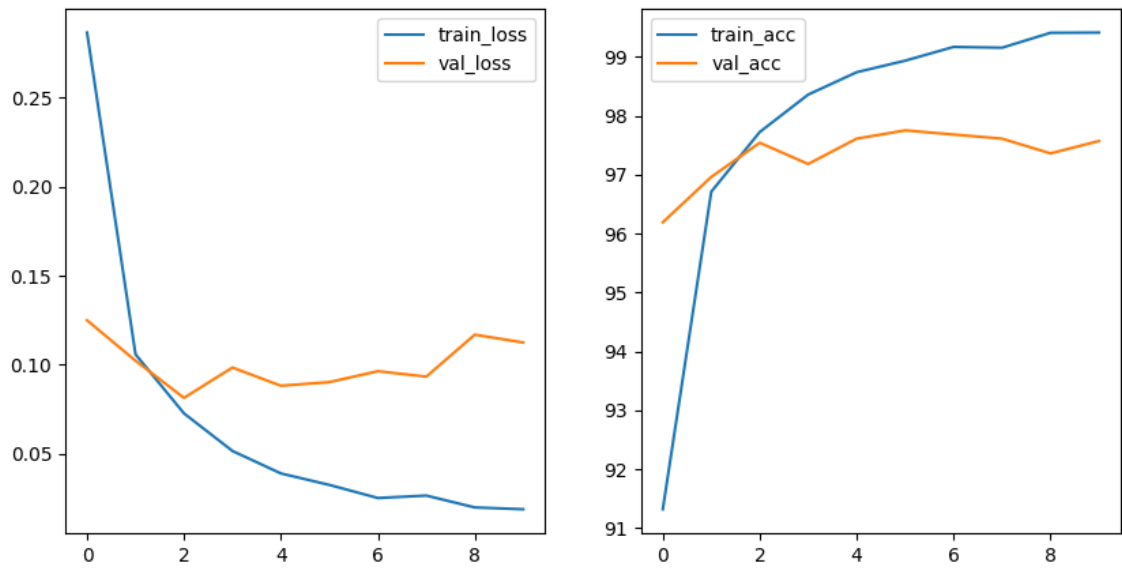
Version 2

使用 Adam 优化器：

```
Train Epoch: 0 [0/50000 (0%)] Loss: 2.316149
Train Epoch: 0 [6400/50000 (13%)] Loss: 0.499585
Train Epoch: 0 [12800/50000 (26%)] Loss: 0.267743
Train Epoch: 0 [19200/50000 (38%)] Loss: 0.359186
Train Epoch: 0 [25600/50000 (51%)] Loss: 0.185033
Train Epoch: 0 [32000/50000 (64%)] Loss: 0.090692
Train Epoch: 0 [38400/50000 (77%)] Loss: 0.168665
Train Epoch: 0 [44800/50000 (90%)] Loss: 0.142781
Train set: Average loss: 0.2865, Accuracy: 45656/50000 (91.31%)
Validation set: Average loss: 0.1250, Accuracy: 9619/10000 (96.19%)
Train Epoch: 1 [0/50000 (0%)] Loss: 0.067316
Train Epoch: 1 [6400/50000 (13%)] Loss: 0.074055
Train Epoch: 1 [12800/50000 (26%)] Loss: 0.129519
Train Epoch: 1 [19200/50000 (38%)] Loss: 0.197798
Train Epoch: 1 [25600/50000 (51%)] Loss: 0.136698
Train Epoch: 1 [32000/50000 (64%)] Loss: 0.087399
Train Epoch: 1 [38400/50000 (77%)] Loss: 0.081349
Train Epoch: 1 [44800/50000 (90%)] Loss: 0.040310
Train set: Average loss: 0.1059, Accuracy: 48353/50000 (96.71%)
Validation set: Average loss: 0.1023, Accuracy: 9696/10000 (96.96%)
Train Epoch: 2 [0/50000 (0%)] Loss: 0.041483
```

```
Train Epoch: 2 [6400/50000 (13%)]      Loss: 0.101126
Train Epoch: 2 [12800/50000 (26%)]     Loss: 0.010579
Train Epoch: 2 [19200/50000 (38%)]     Loss: 0.041791
Train Epoch: 2 [25600/50000 (51%)]     Loss: 0.028775
Train Epoch: 2 [32000/50000 (64%)]     Loss: 0.097586
Train Epoch: 2 [38400/50000 (77%)]     Loss: 0.186257
Train Epoch: 2 [44800/50000 (90%)]     Loss: 0.062413
Train set: Average loss: 0.0727, Accuracy: 48863/50000 (97.73%)
Validation set: Average loss: 0.0814, Accuracy: 9754/10000 (97.54%)
Train Epoch: 3 [0/50000 (0%)]      Loss: 0.010414
Train Epoch: 3 [6400/50000 (13%)]     Loss: 0.044860
Train Epoch: 3 [12800/50000 (26%)]     Loss: 0.014781
Train Epoch: 3 [19200/50000 (38%)]     Loss: 0.055553
Train Epoch: 3 [25600/50000 (51%)]     Loss: 0.012620
Train Epoch: 3 [32000/50000 (64%)]     Loss: 0.021310
Train Epoch: 3 [38400/50000 (77%)]     Loss: 0.103781
Train Epoch: 3 [44800/50000 (90%)]     Loss: 0.041213
Train set: Average loss: 0.0517, Accuracy: 49178/50000 (98.36%)
Validation set: Average loss: 0.0985, Accuracy: 9718/10000 (97.18%)
Train Epoch: 4 [0/50000 (0%)]      Loss: 0.158344
Train Epoch: 4 [6400/50000 (13%)]     Loss: 0.052529
Train Epoch: 4 [12800/50000 (26%)]     Loss: 0.072858
Train Epoch: 4 [19200/50000 (38%)]     Loss: 0.026424
Train Epoch: 4 [25600/50000 (51%)]     Loss: 0.035198
Train Epoch: 4 [32000/50000 (64%)]     Loss: 0.041950
Train Epoch: 4 [38400/50000 (77%)]     Loss: 0.006434
Train Epoch: 4 [44800/50000 (90%)]     Loss: 0.021067
Train set: Average loss: 0.0390, Accuracy: 49368/50000 (98.74%)
Validation set: Average loss: 0.0882, Accuracy: 9761/10000 (97.61%)
Train Epoch: 5 [0/50000 (0%)]      Loss: 0.069585
Train Epoch: 5 [6400/50000 (13%)]     Loss: 0.059508
Train Epoch: 5 [12800/50000 (26%)]     Loss: 0.034162
Train Epoch: 5 [19200/50000 (38%)]     Loss: 0.012449
Train Epoch: 5 [25600/50000 (51%)]     Loss: 0.056644
Train Epoch: 5 [32000/50000 (64%)]     Loss: 0.094050
Train Epoch: 5 [38400/50000 (77%)]     Loss: 0.006455
Train Epoch: 5 [44800/50000 (90%)]     Loss: 0.003520
Train set: Average loss: 0.0326, Accuracy: 49466/50000 (98.93%)
Validation set: Average loss: 0.0903, Accuracy: 9775/10000 (97.75%)
Train Epoch: 6 [0/50000 (0%)]      Loss: 0.026215
Train Epoch: 6 [6400/50000 (13%)]     Loss: 0.009812
Train Epoch: 6 [12800/50000 (26%)]     Loss: 0.078895
Train Epoch: 6 [19200/50000 (38%)]     Loss: 0.004835
Train Epoch: 6 [25600/50000 (51%)]     Loss: 0.001556
Train Epoch: 6 [32000/50000 (64%)]     Loss: 0.015270
Train Epoch: 6 [38400/50000 (77%)]     Loss: 0.021200
```

```
Train Epoch: 6 [38400/50000 (77%)] Loss: 0.031309
Train Epoch: 6 [44800/50000 (90%)] Loss: 0.015980
Train set: Average loss: 0.0252, Accuracy: 49583/50000 (99.17%)
Validation set: Average loss: 0.0964, Accuracy: 9768/10000 (97.68%)
Train Epoch: 7 [0/50000 (0%)] Loss: 0.016455
Train Epoch: 7 [6400/50000 (13%)] Loss: 0.099357
Train Epoch: 7 [12800/50000 (26%)] Loss: 0.000699
Train Epoch: 7 [19200/50000 (38%)] Loss: 0.091024
Train Epoch: 7 [25600/50000 (51%)] Loss: 0.067672
Train Epoch: 7 [32000/50000 (64%)] Loss: 0.067608
Train Epoch: 7 [38400/50000 (77%)] Loss: 0.037591
Train Epoch: 7 [44800/50000 (90%)] Loss: 0.002255
Train set: Average loss: 0.0266, Accuracy: 49576/50000 (99.15%)
Validation set: Average loss: 0.0934, Accuracy: 9761/10000 (97.61%)
Train Epoch: 8 [0/50000 (0%)] Loss: 0.010159
Train Epoch: 8 [6400/50000 (13%)] Loss: 0.006553
Train Epoch: 8 [12800/50000 (26%)] Loss: 0.001041
Train Epoch: 8 [19200/50000 (38%)] Loss: 0.025830
Train Epoch: 8 [25600/50000 (51%)] Loss: 0.001182
Train Epoch: 8 [32000/50000 (64%)] Loss: 0.000722
Train Epoch: 8 [38400/50000 (77%)] Loss: 0.014569
Train Epoch: 8 [44800/50000 (90%)] Loss: 0.005401
Train set: Average loss: 0.0200, Accuracy: 49703/50000 (99.41%)
Validation set: Average loss: 0.1169, Accuracy: 9736/10000 (97.36%)
Train Epoch: 9 [0/50000 (0%)] Loss: 0.003086
Train Epoch: 9 [6400/50000 (13%)] Loss: 0.003649
Train Epoch: 9 [12800/50000 (26%)] Loss: 0.028545
Train Epoch: 9 [19200/50000 (38%)] Loss: 0.003957
Train Epoch: 9 [25600/50000 (51%)] Loss: 0.002014
Train Epoch: 9 [32000/50000 (64%)] Loss: 0.031011
Train Epoch: 9 [38400/50000 (77%)] Loss: 0.044648
Train Epoch: 9 [44800/50000 (90%)] Loss: 0.000788
Train set: Average loss: 0.0189, Accuracy: 49705/50000 (99.41%)
Validation set: Average loss: 0.1125, Accuracy: 9757/10000 (97.57%)
Test set: Accuracy: 9779/10000 (97.79%)
```

分析：将默认的 SGD 替换为 Adam 优化器后，训练速度获得了极其显著的提升。Adam 自带的自适应动量机制有效克服了深层网络梯度微弱与局部鞍点问题，仅需几个 epoch 测试集准确率便轻松突破 97%。同时，可以观察到在最后几个 epoch 中 train accuracy 趋近 100%，而 validation accuracy 无上升趋势，表现出轻微的过拟合现象。

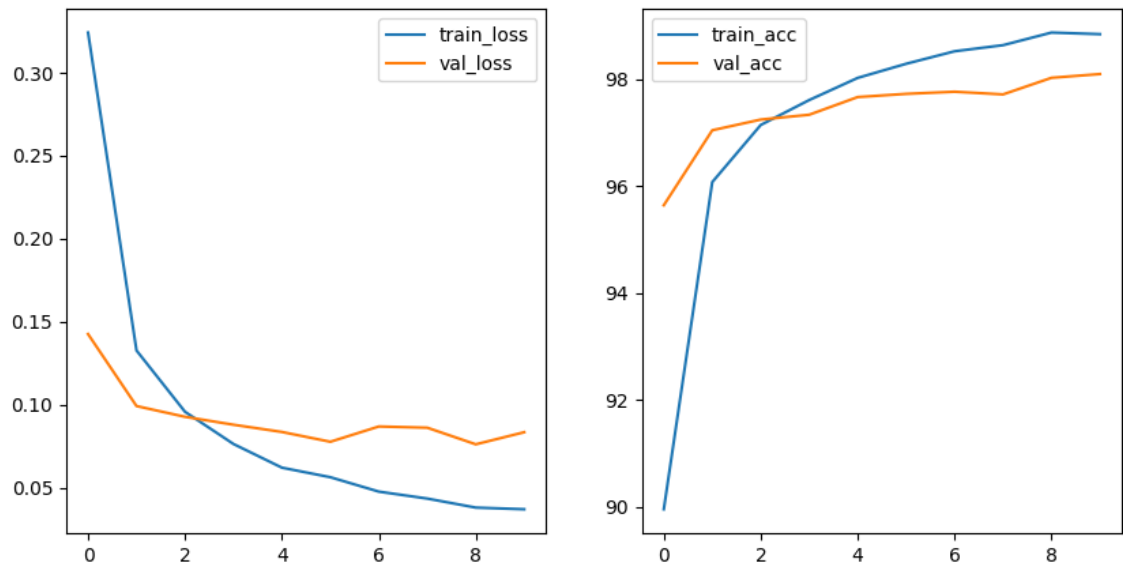
Version 3

添加 Dropout 层：

```
Train Epoch: 0 [0/50000 (0%)]    Loss: 2.304200
Train Epoch: 0 [6400/50000 (13%)]    Loss: 0.555688
Train Epoch: 0 [12800/50000 (26%)]    Loss: 0.525036
Train Epoch: 0 [19200/50000 (38%)]    Loss: 0.261146
Train Epoch: 0 [25600/50000 (51%)]    Loss: 0.191578
Train Epoch: 0 [32000/50000 (64%)]    Loss: 0.242408
Train Epoch: 0 [38400/50000 (77%)]    Loss: 0.411772
Train Epoch: 0 [44800/50000 (90%)]    Loss: 0.083730
Train set: Average loss: 0.3242, Accuracy: 44982/50000 (89.96%)
Validation set: Average loss: 0.1426, Accuracy: 9564/10000 (95.64%)
Train Epoch: 1 [0/50000 (0%)]    Loss: 0.159514
Train Epoch: 1 [6400/50000 (13%)]    Loss: 0.168693
Train Epoch: 1 [12800/50000 (26%)]    Loss: 0.133273
Train Epoch: 1 [19200/50000 (38%)]    Loss: 0.036942
Train Epoch: 1 [25600/50000 (51%)]    Loss: 0.130778
Train Epoch: 1 [32000/50000 (64%)]    Loss: 0.326803
Train Epoch: 1 [38400/50000 (77%)]    Loss: 0.039206
Train Epoch: 1 [44800/50000 (90%)]    Loss: 0.117485
Train set: Average loss: 0.1327, Accuracy: 48034/50000 (96.07%)
Validation set: Average loss: 0.0992, Accuracy: 9704/10000 (97.04%)
Train Epoch: 2 [0/50000 (0%)]    Loss: 0.093282
Train Epoch: 2 [6400/50000 (13%)]    Loss: 0.104371
```

```
Train Epoch: 2 [12800/50000 (26%)]      Loss: 0.178131
Train Epoch: 2 [19200/50000 (38%)]      Loss: 0.121671
Train Epoch: 2 [25600/50000 (51%)]      Loss: 0.163002
Train Epoch: 2 [32000/50000 (64%)]      Loss: 0.008423
Train Epoch: 2 [38400/50000 (77%)]      Loss: 0.063722
Train Epoch: 2 [44800/50000 (90%)]      Loss: 0.035729
Train set: Average loss: 0.0958, Accuracy: 48568/50000 (97.14%)
Validation set: Average loss: 0.0927, Accuracy: 9724/10000 (97.24%)
Train Epoch: 3 [0/50000 (0%)]      Loss: 0.026072
Train Epoch: 3 [6400/50000 (13%)]      Loss: 0.062273
Train Epoch: 3 [12800/50000 (26%)]      Loss: 0.182339
Train Epoch: 3 [19200/50000 (38%)]      Loss: 0.045292
Train Epoch: 3 [25600/50000 (51%)]      Loss: 0.061666
Train Epoch: 3 [32000/50000 (64%)]      Loss: 0.038512
Train Epoch: 3 [38400/50000 (77%)]      Loss: 0.243312
Train Epoch: 3 [44800/50000 (90%)]      Loss: 0.171799
Train set: Average loss: 0.0764, Accuracy: 48807/50000 (97.61%)
Validation set: Average loss: 0.0880, Accuracy: 9733/10000 (97.33%)
Train Epoch: 4 [0/50000 (0%)]      Loss: 0.061891
Train Epoch: 4 [6400/50000 (13%)]      Loss: 0.012929
Train Epoch: 4 [12800/50000 (26%)]      Loss: 0.052093
Train Epoch: 4 [19200/50000 (38%)]      Loss: 0.075200
Train Epoch: 4 [25600/50000 (51%)]      Loss: 0.070733
Train Epoch: 4 [32000/50000 (64%)]      Loss: 0.011389
Train Epoch: 4 [38400/50000 (77%)]      Loss: 0.022270
Train Epoch: 4 [44800/50000 (90%)]      Loss: 0.006259
Train set: Average loss: 0.0621, Accuracy: 49008/50000 (98.02%)
Validation set: Average loss: 0.0836, Accuracy: 9766/10000 (97.66%)
Train Epoch: 5 [0/50000 (0%)]      Loss: 0.059957
Train Epoch: 5 [6400/50000 (13%)]      Loss: 0.026749
Train Epoch: 5 [12800/50000 (26%)]      Loss: 0.144747
Train Epoch: 5 [19200/50000 (38%)]      Loss: 0.038615
Train Epoch: 5 [25600/50000 (51%)]      Loss: 0.014902
Train Epoch: 5 [32000/50000 (64%)]      Loss: 0.015914
Train Epoch: 5 [38400/50000 (77%)]      Loss: 0.140517
Train Epoch: 5 [44800/50000 (90%)]      Loss: 0.025031
Train set: Average loss: 0.0563, Accuracy: 49140/50000 (98.28%)
Validation set: Average loss: 0.0777, Accuracy: 9772/10000 (97.72%)
Train Epoch: 6 [0/50000 (0%)]      Loss: 0.132602
Train Epoch: 6 [6400/50000 (13%)]      Loss: 0.064819
Train Epoch: 6 [12800/50000 (26%)]      Loss: 0.108420
Train Epoch: 6 [19200/50000 (38%)]      Loss: 0.088527
Train Epoch: 6 [25600/50000 (51%)]      Loss: 0.008238
Train Epoch: 6 [32000/50000 (64%)]      Loss: 0.065598
Train Epoch: 6 [38400/50000 (77%)]      Loss: 0.037360
Train Epoch: 6 [44800/50000 (90%)]      Loss: 0.072200
```

```
Train Epoch: 6 [44800/50000 (90%)] Loss: 0.072202
Train set: Average loss: 0.0477, Accuracy: 49257/50000 (98.51%)
Validation set: Average loss: 0.0869, Accuracy: 9776/10000 (97.76%)
Train Epoch: 7 [0/50000 (0%)] Loss: 0.027459
Train Epoch: 7 [6400/50000 (13%)] Loss: 0.003749
Train Epoch: 7 [12800/50000 (26%)] Loss: 0.020759
Train Epoch: 7 [19200/50000 (38%)] Loss: 0.118453
Train Epoch: 7 [25600/50000 (51%)] Loss: 0.000506
Train Epoch: 7 [32000/50000 (64%)] Loss: 0.003093
Train Epoch: 7 [38400/50000 (77%)] Loss: 0.108960
Train Epoch: 7 [44800/50000 (90%)] Loss: 0.046219
Train set: Average loss: 0.0435, Accuracy: 49314/50000 (98.63%)
Validation set: Average loss: 0.0862, Accuracy: 9771/10000 (97.71%)
Train Epoch: 8 [0/50000 (0%)] Loss: 0.044992
Train Epoch: 8 [6400/50000 (13%)] Loss: 0.058512
Train Epoch: 8 [12800/50000 (26%)] Loss: 0.097902
Train Epoch: 8 [19200/50000 (38%)] Loss: 0.116848
Train Epoch: 8 [25600/50000 (51%)] Loss: 0.032235
Train Epoch: 8 [32000/50000 (64%)] Loss: 0.051297
Train Epoch: 8 [38400/50000 (77%)] Loss: 0.063546
Train Epoch: 8 [44800/50000 (90%)] Loss: 0.011301
Train set: Average loss: 0.0380, Accuracy: 49432/50000 (98.86%)
Validation set: Average loss: 0.0762, Accuracy: 9802/10000 (98.02%)
Train Epoch: 9 [0/50000 (0%)] Loss: 0.006851
Train Epoch: 9 [6400/50000 (13%)] Loss: 0.001762
Train Epoch: 9 [12800/50000 (26%)] Loss: 0.026911
Train Epoch: 9 [19200/50000 (38%)] Loss: 0.058147
Train Epoch: 9 [25600/50000 (51%)] Loss: 0.007830
Train Epoch: 9 [32000/50000 (64%)] Loss: 0.006466
Train Epoch: 9 [38400/50000 (77%)] Loss: 0.139453
Train Epoch: 9 [44800/50000 (90%)] Loss: 0.074731
Train set: Average loss: 0.0370, Accuracy: 49417/50000 (98.83%)
Validation set: Average loss: 0.0834, Accuracy: 9809/10000 (98.09%)
Test set: Accuracy: 9823/10000 (98.23%)
```

分析：在每层隐藏层激后添加 Dropout(0.2) 后，通过让神经元随机失活，迫使全连接网络避免对局部特征的严重同质化依赖。虽然牺牲了少量虚高的 train accuracy，但有效缩小了泛化差距，减少了过拟合。

最终采用 Version 3